LC Gate Serial Comm Data Format

Note: All the bytes are in INTEGER format NOT ASCII

From Controller to the Tablet :

Byte 0-1: Header (0x55)	**Byte 2:** Train UP	**Byte 3:** Train Down	**Byte 4:** Boom1 Status	**Byte 5:** Boom2 Status	**Byte 6:** Boom1 Health	**Byte 7:** Boom 2 Health	**Byte 8:** Lock Status	**Byte 9:** Alarm Status	**Byte 10:** Bypass Status

- **Byte 0 - 1: Header (0x55)** - First two lower bytes of the data is the Header , which basically is used to check if the communicated buffer received is correct.

- **Byte 2: Train UP** - This byte is set to 0x01 - when a train is being detected in this UP Track , set to 0x00 - when no train is being detected , and set to 0x11 - when either of the two sensors fails to communicate.

- **Byte 3: Train Down** - This byte is set to 0x01 - when a train is being detected in this DOWN Track , set to 0x00 - when no train is being detected , and set to 0x11 - when either of the two sensors fails to communicate.

- **Byte 4: Boom1 Status** - This byte is set to 0x01 - when the first boom is UP and set to 0x00 - when the boom is DOWN.

- **Byte 5: Boom2 Status** - This byte is set to 0x01 - when the first boom is UP and set to 0x00 - when the boom is DOWN.

- **Byte 6: Boom1 Health** - This byte is set to 0x01 - when boom1 is healthy and is set 0x00 - when not healthy.

- **Byte 7: Boom2 Health** - This byte is set to 0x01 - when boom2 is healthy and is set 0x00 - when not healthy.

- **<u>Byte 8: </u>Lock Status** - This Byte is set to 0x01 when Locked and its set to 0x00 when Unlocked.

- **<u>Byte 9: Alarm Status</u>** - This byte is set to 0x01 - when Alarm is ON. and set to 0x00 when alarm is OFF.

- **<u>Byte 10: Lever Status</u>** - This byte is set to 0x01 - when Bypassed and set to 0x00 when lock is not bypassed.Set to 0x11 when bypass switch fails , 0x0F if Breached.

<u>From Tablet to the Controller </u>:

<u>Byte 0-1:</u> Header (0x59)	**<u>Byte 2:</u>** Lever Lock	**<u>Byte 3:</u>** Alarm ON	**<u>Byte 4:</u>** Send Data

- **<u>Byte 0-1: Header (0x55)</u>** - Used to check if the received buffer is properly aligned and correct.

- **<u>Byte 2: Lever Lock</u>** - Set to 0x00 - to turn on the Coil in the EKT which lets the GK to unlock the key. Locking setup is done by GK manually…

- **<u>Byte 3: Alarm ON</u>** - Set to 0x01 - To turn ON the Alarm and set to 0x00 - to turn OFF the Alarm.

- **<u>Byte 4: Send Data</u>** - Set to 0x01 - To request for the sensor data. This command sends all the sensor data (11 byte) to the device.

Note: Make the bytes 0xFF to be ignored , so that you don't have to send something again and again for no reason.